



UNIVERSITY OF NAIROBI

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**DEPARTMENT OF MECHANICAL AND
MANUFACTURING ENGINEERING**

FINAL YEAR PROJECT PROPOSAL

**Bridging the Gap: Computational Prediction vs. Physical
Performance of 3D-Printed Lattice Structures**

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Bachelor of Science in Mechanical and Manufacturing Engineering*

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1. Background

The relentless demand for efficiency and high performance in modern engineering sectors, particularly aerospace and automotive, has driven the pursuit of lightweight structural materials. Historically, weight reduction involved removing bulk material, which often compromised structural integrity. The advent of Additive Manufacturing (AM) has fundamentally shifted this paradigm, enabling the fabrication of complex, bio-inspired architected cellular materials, known as lattice structures, which offer exceptional strength-to-weight ratios and energy absorption capabilities.

While topological research into mesoscale lattices demonstrates that they can achieve high-level performance comparable to traditional solid metals at a fraction of the weight, the transition from digital design to physical application presents significant engineering challenges. 3D-printed materials exhibit anisotropic behavior, and their real-world mechanical properties are heavily influenced by manufacturing parameters such as layer adhesion, print orientation, and micro-voids. Consequently, an analytical gap exists between idealized computational models and the empirical performance of fabricated components. Closing this loop through rigorous physical testing is paramount for the reliable application of additive manufacturing in load-bearing environments.

2. Problem Statement

Finite Element Analysis (FEA) is widely utilized to predict the mechanical behavior of complex lattice structures; however, these computational models often rely on bulk isotropic material properties that do not accurately represent the realities of Fused Deposition Modeling (FDM) 3D printing. As a result, simulations frequently overestimate the stiffness and yield strength of printed components, leading to potential catastrophic failures in physical applications. There is a critical need to characterize the exact deviation between idealized computational predictions and the actual mechanical behavior of 3D-printed lattices. By establishing a direct correlation between FEA simulations and physical compression testing, engineers can develop more accurate predictive models and optimize design-for-manufacturing guidelines for lightweight structural components.

3. Aim

The aim of this project is to design, simulate, fabricate, and test 3D-printed lattice structures for lightweight structural applications, and to validate finite element analysis predictions using experimental compression testing.

4. Objectives

1. To design parametric lattice structures using Autodesk Fusion.
2. To perform hand calculations for mass, relative density, and estimated stiffness.
3. To simulate the compression behavior of lattice structures using finite element analysis.
4. To fabricate selected lattice specimens using fused deposition modeling (FDM) 3D printing.
5. To conduct quasi-static compression tests and obtain empirical force-displacement curves.
6. To compare simulation results with experimental results and discuss sources of variance and error.

5. Scope

This project is confined to the structural analysis of polymer-based lattice topologies (specifically Simple Cubic and Body-Centered Cubic structures) subjected to quasi-static axial compression. The physical specimens will be fabricated from Polylactic Acid (PLA) utilizing FDM technology. The scale of the test specimens will be restricted to standard coupon sizes (e.g., 60 mm $\times$ 60 mm $\times$ 60 mm) featuring integrated solid end-caps to ensure uniform load distribution during testing. The scope strictly excludes dynamic impact testing, thermal analysis, and the use of metal additive manufacturing.

6. Methodology

- **Analytical Modeling & CAD Generation:** Parametric models of the lattice unit cells and full arrays will be generated using Autodesk Fusion. Theoretical parameters, including relative density and estimated stiffness, will be mathematically derived using the Gibson-Ashby model for cellular solids to establish a baseline for comparison.
- **Computational Simulation (FEA):** The CAD models will be subjected to linear static compression studies within the Autodesk Fusion Simulation workspace or SimScale. A rigorous mesh convergence study will be executed to ensure numerical accuracy, yielding predicted values for maximum displacement, von Mises stress distribution, and theoretical stiffness.

- **Fabrication:** The optimized lattice designs will be exported as STL files and processed through Ultimaker Cura. Physical prototypes will be fabricated using an FDM 3D printer at the university workshops or FabLab Nairobi. Printing parameters (layer height, infill density, extrusion temperature) will be tightly controlled and documented.
- **Experimental Testing:** Dimensional inspection and mass measurements will be conducted post-fabrication. The specimens will then be subjected to quasi-static axial compression using a Universal Testing Machine (UTM) at a controlled displacement rate until structural failure or a predetermined strain limit is reached.
- **Validation & Analysis:** Force-displacement data logged by the UTM will be converted into engineering stress-strain curves. The empirical specific stiffness and failure modes will be directly compared against the FEA predictions. The variance will be calculated, and extensive error analysis will be conducted to explain discrepancies caused by manufacturing realities.

7. Expected Results

- Fully parameterized CAD models and professional engineering drawings of multiple lattice topologies.
- Validated computational simulation reports demonstrating stress concentrations and mesh convergence.
- Empirical force-displacement and stress-strain curves highlighting the physical yield strength and energy absorption of the tested structures.
- A quantified error metric detailing the percentage difference between simulated stiffness and physical stiffness.
- A comprehensive final report providing actionable design-for-manufacturing (DfAM) recommendations for lightweight polymer lattices.

8. Tools and Resources

- **Software:** Autodesk Fusion (CAD Modeling and FEA), SimScale (Secondary Cloud FEA), Ultimaker Cura (Slicing Software), Microsoft Excel (Data Processing).
- **Hardware/Equipment:** FDM 3D Printer, Universal Testing Machine (UTM), Digital Calipers, Digital Weighing Scale.
- **Materials:** PLA (Polylactic Acid) 3D printing filament.

9. Timeline

- **Weeks 1–3:** Literature review, analytical Gibson-Ashby hand calculations, and parametric CAD modeling of lattice unit cells.
- **Weeks 4–6:** FEA setup, material property assignment, mesh convergence studies, and extraction of theoretical stiffness data.
- **Weeks 7–9:** Generation of G-code, 3D printing of calibration cubes and final lattice specimens, followed by dimensional and mass inspection.
- **Weeks 10–12:** Execution of experimental compression testing on the Universal Testing Machine; recording of failure modes and raw data extraction.
- **Weeks 13–16:** Data processing (stress-strain conversion), comparative analysis between physical and simulated results, drafting of the final capstone report, and defense presentation preparation.

10. References

- [1] Gibson, L. J., & Ashby, M. F. (1997). *Cellular Solids: Structure and Properties* (2nd ed.). Cambridge University Press.
- [2] ASTM International. (2015). *ASTM D1621-16 Standard Test Method for Compressive Properties of Rigid Cellular Plastics*.
- [3] Maconachie, T., Leary, M., Lozanovski, B., Zhang, X., Qian, M., Faruque, O., & Brandt, M. (2019). SLM lattice structures: Properties, performance, applications and challenges. *Materials & Design*, 183, 108137.
- [4] Tamburrino, F., Graziosi, S., & Bordegoni, M. (2018). The design process of additively manufactured mesoscale lattice structures: A review. *Journal of Computing and Information Science in Engineering*, 18(4), 040801.